

GRYTVIKEN, South Georgia and the South Sandwich Islands

WMO: 889030

Lat: 54.284S

Lon: 36.496W

Elev: 10

StdP: 14.69

Time Zone: -2.00 (W02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	18.8	4.7	7.0	22.4	7.8	8.2	23.0	33.4	32.5	30.2	36.1	5.5	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.2	60.4	50.0	57.2	48.4	54.2	46.1	51.3	58.3	49.0	55.1	46.7	52.6	18.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
46.6	47.0	54.5	44.3	42.9	52.2	42.4	40.0	49.6	21.1	58.7	19.8	55.3	18.6	52.8	61.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.0	28.3	25.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	42.8	44.2	41.2	37.2	34.1	31.4	30.0	29.9	33.5	36.8	39.5	41.8
	DBStd	7.08	4.17	5.03	4.38	4.99	6.36	5.43	5.99	5.69	5.41	4.89	4.46	4.56
	HDD50	4837	227	176	275	384	495	558	619	623	495	411	314	260
	HDD65	10283	687	582	737	834	958	1008	1084	1088	945	875	764	721
	CDD50	30	5	15	3	0	2	0	0	0	0	0	1	4
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	10.3	10.5	11.8	10.9	9.6	9.7	9.1	9.7	10.2	10.9	10.3	10.7	10.9
	PrecAvg	64.8	4.7	5.1	6.9	6.3	6.6	5.7	5.7	5.9	5.5	4.8	4.1	3.6
(15) Precipitation	PrecMax	77.6	7.6	6.8	9.8	10.1	9.9	8.0	9.4	9.5	12.2	8.3	6.0	5.2
	PrecMin	50.7	2.7	2.4	5.3	4.1	4.4	3.7	2.5	4.1	0.9	2.6	1.9	2.0
	PrecStd	7.8	1.3	1.1	1.2	1.3	1.1	1.1	1.3	1.3	1.8	1.1	0.8	0.9
	PrecStd	7.8	1.3	1.1	1.2	1.3	1.1	1.1	1.3	1.3	1.8	1.1	0.8	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	64.0	66.2	61.8	57.3	57.2	50.9	53.5	48.3	50.7	56.6	58.9	61.8	
	0.4% MCWB	52.2	55.1	50.4	48.1	49.5	40.9	44.1	41.6	41.9	47.5	47.1	49.4	
	2% DB	56.9	60.9	56.5	53.5	51.9	45.9	45.3	44.7	47.0	51.8	54.0	55.9	
	2% MCWB	47.4	50.9	48.3	45.9	44.8	38.4	38.5	38.3	39.9	41.7	46.3	46.4	
	5% DB	53.1	57.3	52.3	48.5	47.0	41.8	41.9	41.8	44.3	48.2	50.2	52.9	
	5% MCWB	44.5	49.6	45.9	42.0	41.0	36.3	36.5	36.3	37.7	40.9	42.6	45.0	
	10% DB	49.9	53.3	49.1	44.7	43.1	39.1	38.9	39.1	41.8	45.1	47.6	50.1	
	10% MCWB	43.0	46.3	43.6	40.2	37.7	34.3	34.4	34.6	36.3	38.9	40.8	42.9	
	0.4% WB	54.0	57.0	53.0	50.2	49.2	43.8	43.1	42.2	42.5	46.9	50.1	49.5	
	0.4% MCDB	63.3	65.9	60.9	56.4	56.1	49.6	47.9	47.3	48.3	52.9	55.6	55.8	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	48.4	52.3	48.7	46.5	45.0	39.0	39.3	39.3	40.4	42.9	46.6	47.1	
	2% MCDB	54.9	59.7	54.1	53.0	50.5	44.1	44.3	43.3	46.0	48.2	51.9	53.4	
	5% WB	45.4	49.6	46.4	43.7	41.2	36.8	36.9	36.7	38.5	40.8	43.7	45.0	
	5% MCDB	51.0	56.0	51.5	48.4	46.5	41.3	41.1	40.7	43.3	46.3	49.1	51.4	
	10% WB	43.6	46.5	44.0	40.9	38.2	34.8	34.8	34.9	36.7	39.1	41.7	43.1	
	10% MCDB	48.7	52.6	48.4	44.1	42.4	38.3	38.3	38.5	40.9	43.9	46.3	49.4	
(35) Mean Daily Temperature Range	MDBR	11.4	12.2	11.4	10.7	8.8	8.2	8.3	9.1	9.9	12.1	11.1	12.3	
	5% DB MCDBR	17.1	19.3	18.7	16.9	14.6	13.9	12.3	12.7	13.8	17.3	16.8	17.7	
	5% DB MCWBR	10.3	12.2	12.7	12.5	10.2	9.7	9.3	9.9	10.0	11.6	10.7	11.1	
	5% WB MCDBR	15.4	18.7	17.3	16.5	13.1	12.6	11.4	12.0	13.4	15.9	15.1	17.4	
	5% WB MCWBR	10.4	12.6	12.9	12.9	10.4	9.5	9.4	9.8	10.5	11.8	11.7	12.0	
	5% WB MCWBR	10.4	12.6	12.9	12.9	10.4	9.5	9.4	9.8	10.5	11.8	11.7	12.0	
(40) Clear-Sky Solar Irradiance	taub	0.334	0.331	0.324	0.306	0.281	0.268	0.280	0.299	0.315	0.324	0.321	0.327	
	taud	2.492	2.506	2.511	2.479	2.363	2.260	2.303	2.373	2.394	2.403	2.459	2.485	
	Ebn at Noon	306	295	276	245	206	180	200	241	273	293	309	312	
	Edh at Noon	33	30	26	21	16	14	16	22	29	33	34	34	
(44) All-Sky Solar Radiation	RadAvg	1348	1087	710	391	190	121	152	303	599	946	1335	1512	
	RadStd	86	75	48	35	16	8	13	21	50	55	103	93	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page